## 0.1 Why the Response Remains of the Closed Form: Proposition, Bound, and Verification

### 0.1.1 Exact dynamics and the structure of the perturbations

About the active pivot line $P$, the exact roll dynamics during ground contact read (the pitch case is analogous)

$$J_P\,\dot{\omega} = M_{x,P}(t) + \Delta M_{\mathrm{GE},P}(t,\phi) - W(l_p + \lambda_{\mathrm{off}})\cos\phi + W z_{\mathrm{CoM}}\sin\phi, \tag{1}$$

with $M_{x,P} = M_x + f\,l_p$ and, conservatively, $l_{p,\phi} = 0.140$ m, $l_{p,\theta} = 0.110$ m. The ground-contact phase is a one-degree-of-freedom rigid rotation about the gear line, so every rotor height is an exact trigonometric function of the single angle $\phi$; in the level configuration all heights coincide, yielding the exact identity

$$\Delta M_{\mathrm{GE},P} = \eta(\phi)\,\big(M_x + f l_p\big), \qquad \eta(\phi) \triangleq \bar{\gamma}(\phi) - 1, \tag{2}$$

where $\bar{\gamma}$ denotes the rotor-common thrust factor. The identity is *independent of the CoM position* — $\lambda_{\mathrm{off}}$ enters (1) only through the gravity arm — so ground effect biases the identified critical moment identically for every CoM configuration. The magnitude is evaluated with two models of increasing physics content. The per-rotor superposition of the single-rotor Cheeseman–Bennett result (which *neglects* rotor–rotor interference) gives, at the measured geometry ($h = 0.315$ m, $R = 0.127$ m), $\eta(0) = 1.03\%$ independent of $l_p$. The reported model adds the interference from first principles by the image method: each rotor is a three-dimensional potential source of strength $R^2 v_\infty/4$, the ground plane is enforced by mirror images, and only the vertical projection of each image-induced velocity reduces the downwash, giving the per-rotor sum

$$s_{\mathrm{rot},i} \;=\; \frac{R^2}{4}\sum_j \frac{z_i + z_j}{\big[d_{ij}^2 + (z_i + z_j)^2\big]^{3/2}}, \qquad \gamma_i = \big(1 - s_{\mathrm{rot},i}\big)^{-1}, \tag{3}$$

evaluated at each rotor's *own* tilted height $z_i = h\cos\phi + s_i\sin\phi$ — no averaging enters, and the attitude dependence of the model resides entirely in this term. The self term $j = i$ reduces exactly to Cheeseman–Bennett, so (3) contains the single-rotor result as the zero-neighbour limit and adds *no empirical constant of any kind*: the neighbour sum is fixed by the measured rotor geometry alone. Mapping the per-rotor gains to the pivot,

$$\Delta M_{\mathrm{GE}} = \sum_i b_i\,(\gamma_i - 1)\,T_i \;+\; \Big[\sum_i (\gamma_i - 1)\,T_i\Big] l_p, \tag{4}$$

with $b_i$ the body-centre arms and the second term the parallel-axis transfer of the net vertical surplus. Under constant total thrust the result is affine in the commanded moment, $\Delta M_{\mathrm{GE}} = a(\phi) + b(\phi) M_{\mathrm{cmd}}$, with fitted coefficients $a/(f l_p)\,[\%] = 4.27 - 0.086\,\phi$ and $b\,[\%] = 4.28 - 0.060\,\phi$ ($\phi$ in degrees) — a factor $\approx 4$ above the interference-free value in both channels, which is the quantitative statement that rotor–rotor interference is not negligible at this clearance. The affine decomposition, the anti-restoring sign of the height-asymmetry moment, and the channel structure below are identical in the two models — they follow from thrust-proportionality and smooth height dependence alone; only the magnitudes differ.

How far does this move the calibrated constants? With $k_0^{\mathrm{GE}} \triangleq \beta_f f l_p + \beta_M M_{\mathrm{crit}}$ the onset value of the $\phi$-slope, the effective anti-restoring gradient becomes $D = W z_{\mathrm{CoM}} + k_0^{\mathrm{GE}}$ ($k_0^{\mathrm{GE}} < 0$: the ground effect *opposes* the tilt and therefore reduces $D$). Since $C_2 = \sqrt{D/J_P}$, the inertia cancels in the ratio and no inertia estimate is needed:

$$\frac{\Delta C_2}{C_2} = \sqrt{1 - \frac{|k_0^{\mathrm{GE}}|}{W z_{\mathrm{CoM}}}} - 1, \qquad \frac{\Delta K}{K} = \frac{1 + \alpha_M}{1 - |k_0^{\mathrm{GE}}|/(W z_{\mathrm{CoM}})} - 1. \tag{5}$$

Over the operating box ($W z_{\mathrm{CoM}} \in [6.3, 9.5]$ N m for $z_{\mathrm{CoM}} \in [0.2, 0.3]$ m, measured onset thrust $f \approx 21$ N, $|M_{\mathrm{crit}}| \le 2.1$ N m) the interference model gives $|k_0^{\mathrm{GE}}| \le 0.217$ N m/rad (roll) and 0.172 (pitch), i.e. at most 3.4% and 2.7% of $W z_{\mathrm{CoM}}$, so

$$\frac{\Delta C_2}{C_2} \ge -1.7\%\ \text{(roll)}, \quad -1.4\%\ \text{(pitch)}, \qquad \frac{\Delta K}{K} \in [+6.2\%, +8.0\%],$$

the amplitude constant moving more because it carries the two effects multiplicatively, $1.0428 \times (0.96566)^{-1}$: the moment-channel gain $\alpha_M$ raises the effective ramp rate while the gradient reduction raises $1/D$, whereas $C_2$ receives only the latter and under a square root. The $K$ shift is one-signed over the whole box, so neglecting ground effect does not merely blur the amplitude constant — it underestimates it by at least 6%. Neither shift is an error. The calibration estimates the *effective* constants directly, so $\alpha_M$

and $k_0^{\mathrm{GE}}$ are absorbed by construction — provided each is common to the runs that share one calibrated pair. Both are, to the measured precision: the moment-channel gain is rate-invariant over the twelvefold rate range (IQR $[4.13, 4.27]$%, a 0.14 pp spread, i.e. 0.13% of $K$), and within one dataset the only run-to-run variation of $k_0^{\mathrm{GE}}$ enters through $M_{\mathrm{crit}}$, whose largest directional spread (0.96 N m) changes $D$ by 0.52% and $C_2$ by 0.26% — so a *single* $C_2$ per configuration remains legitimate, and only those residual spreads escape absorption. What the shifts do say is that the calibrated constants are not their textbook counterparts: $1/K$ underestimates $W z_{\mathrm{CoM}}$ by 6–8% and must not be reported as a measurement of $z_{\mathrm{CoM}}$. The ±20% sensitivity result — a mis-specification of either constant moves the identified CoM offset by less than 0.9 mm — is the fallback for a user who pins the constants a priori instead of calibrating; it is not what protects the absorbed shifts. For scale, the a priori uncertainty of $C_2$ from the $z_{\mathrm{CoM}}$ box alone is $-12.5\%$ ($6.14 \to 5.37$ rad/s with $J_P \geq m(z_{\mathrm{CoM}}^2 + l_p^2)$), several times the ground-effect shift — which is precisely why $(C_2, K)$ are calibrated rather than computed.

Everything that follows is deliberately insensitive to that magnitude. The absorption argument assumes only smoothness and thrust-proportionality, which any near-ground aerodynamic model shares, and the same is true of a purely geometric alternative: a contact-force lever displaced inboard of the kinematic pivot circle by a finite, compliant contact patch enters (1) in the same functional position as the $a$-channel. The static check of the identified thresholds indeed cannot separate the two — a $19.2 \pm 1.8$ mm inboard lever reproduces the measured deficit as well as the interference model does (residual RMS 82.3 versus 86.0 mN m), the discriminating moment-proportional channel being below the group-to-group scatter. This is a limitation of the attribution, not of the method: the bounds below hold for either cause, and both cancel antisymmetrically in the pivot-free deliverable.

### 0.1.2 Ideal configuration: the cosh family and its escape channel

Consider first the ideal level configuration: the vehicle rests at $\phi = 0$, so the onset — defined dynamically by zero net moment, $\omega(0) = \dot{\omega}(0) = 0$ — occurs *at* $\phi = 0$, and the operating point $(\tau, \phi, T) = (0, 0, T^*)$ makes the onset expansion and the textbook small-angle expansion one and the same. The tilt variable and the excursion coincide, $\delta\phi = \phi$.

Linearizing (1) about this point collects every perturbation into three absorbable channels plus a remainder: constants are absorbed by the onset anchoring; terms $\propto \dot{M}\tau$ (including the thrust-ramp channel of (2), whose coefficient $\eta^*$ is rate-independent) are absorbed by the ramp-invariance calibration of $K$; terms $\propto \delta\phi$ (the gravity slope $W z_{\mathrm{CoM}}$ — here an *anti-restoring* gradient, i.e. a negative stiffness, which is what makes the response hyperbolic rather than oscillatory — and $\eta' M_P^*$) are absorbed by the calibrated $C_2$. What escapes is of two kinds — the linearization remainder of the gravity terms and, from the ground effect, the one product that is bilinear in two spanned coordinates. Subtracting the tangent $G_{\mathrm{lin}} = -Wa + W z_{\mathrm{CoM}}\,\delta\phi$ from the exact gravity moment leaves the remainder *in closed form* — no Lagrange point $\xi$ and no truncation:

$$\rho \;=\; \underbrace{Wa\,(1-\cos\delta\phi) \;+\; W z_{\mathrm{CoM}}\,(\sin\delta\phi - \delta\phi)}_{\rho_\phi,\ \text{gravity (exact)}} \;+\; \underbrace{\beta_M\,\dot{M}\,\tau\,\delta\phi}_{\rho_b,\ \text{ground effect (bilinear)}} \;+\;\ldots, \qquad a \triangleq l_p + \lambda_{\mathrm{off}}, \quad (6)$$

the second term arising because the moment-proportional ground-effect coefficient is itself attitude dependent: writing $b(\phi) = \alpha_M + \beta_M\delta\phi$ and $M_{\mathrm{cmd}} = M_{\mathrm{crit}}^* + \dot{M}\tau$, the product $b(\phi)M_{\mathrm{cmd}}$ splits into a constant, a term $\propto \dot{M}\tau$, a term $\propto \delta\phi$ with coefficient $\beta_M M_{\mathrm{crit}}^*$ — all three absorbed above — and this single bilinear leftover. It is quantified in Sec. 0.1.3 and enters Table 2 on its own row.

In $\rho_\phi$ the two terms have opposite signs and different orders: the arm term is $+\frac{1}{2}Wa\,\delta\phi^2$ to leading order, the $z_{\mathrm{CoM}}$ term $-\frac{1}{6}W z_{\mathrm{CoM}}\,\delta\phi^3$, so dropping the latter recovers the familiar quadratic bound $\frac{1}{2}Wa\,\delta\phi^2$ conservatively. Because that third-order term carries the only $z_{\mathrm{CoM}}$ dependence, the whole $z_{\mathrm{CoM}} \in [0.2, 0.3]$ m box moves $\rho_\phi$ by less than 4%; the worst case is the *smallest* $z_{\mathrm{CoM}}$. Evaluated with $W = 31.59$ N, $a = l_p + 0.020$ m and $z_{\mathrm{CoM}} = 0.2$ m, Table 1 gives the resulting envelope at the design cap and at the largest absolute tilt reached.

Table 1: Exact gravity remainder $\rho_\phi$ [mN m] from (6), at $z_{\mathrm{CoM}} = 0.2$ m (worst case).

| | $a$ [m] | $\delta\phi = 5^\circ$ (design cap) | $\delta\phi = 10^\circ$ (largest absolute tilt) |
|---|---|---|---|
| Roll | 0.160 | 18.5 | 71.2 |
| Pitch | 0.130 | 14.9 | 56.8 |

**Proposition 1** *Under (i) maintained pivot contact, (ii) gate-passing ramp execution, and (iii) any thrust-proportional, smooth ground-effect model, subtracting the nominal (onset-balanced) equation from* (1) *yields for the deviation* $e_\phi \triangleq \phi - \phi_{\text{nom}}$

$$J_P\, \ddot{e}_\phi - W z_{\text{CoM}}\, e_\phi = \rho(\tau), \qquad e_\phi(0) = \dot{e}_\phi(0) = 0, \tag{7}$$

*where the state-dependent remainder is recast as an equivalent exogenous input,* $\rho(\tau) \triangleq \rho(\delta\phi(\tau))$*, and* $W z_{\text{CoM}} = J_P C_2^2$ *is understood as the* effective *anti-restoring gradient that the calibrated* $C_2$ *estimates. Every solution of the unforced problem with the onset conditions* $\omega(0) = \dot{\omega}(0) = 0$ *is exactly* $\omega(\tau) = C_1(\cosh C_2\tau - 1)$ *with* $C_1 = K\dot{M}$ *fixed — no shape parameter is free. Ground effect and the gravity nonlinearity therefore move the system* within *the two-parameter cosh family; the only escape channel is* $\rho$*.*

**Impulse response and the propagated bound.** The response of (7) to a unit moment impulse $\rho = \delta(\tau)$ follows from the momentum jump $J_P[\dot{e}_\phi(0^+) - \dot{e}_\phi(0^-)] = 1$ (the position cannot jump) and the unstable homogeneous solution: $e_\phi = \sinh(C_2\tau)/(J_P C_2)$, hence the rate impulse response $h_\omega(\tau) = \dot{e}_\phi = \cosh(C_2\tau)/J_P$ — consistent with $h_\omega(0) = 1/J_P$ from the jump condition. Superposition (Duhamel's integral [*ref*]) then gives the exact solution for arbitrary $\rho$ and, normalizing like the empirical fit metric by the window-edge signal, the finite monotone envelope

$$\delta\omega(\tau) = \frac{1}{J_P}\int_0^\tau \cosh\big(C_2(\tau - s)\big)\rho(s)\, ds, \qquad \frac{|\delta\omega(\tau)|}{\omega_{\text{nom}}(\tau_{\text{end}})} \leq \frac{C_2\, \rho_{\max}}{\dot{M}}\, \frac{\sinh(C_2\tau)}{\cosh(C_2\tau_{\text{end}}) - 1}, \tag{8}$$

with $\rho_{\max} \triangleq \sup_{[0,\tau_{\text{end}}]} |\rho|$, attaining $(C_2\rho_{\max}/\dot{M})\coth(C_2\tau_{\text{end}}/2)$ at the edge. Three points sharpen the bound. (a) *Kernel–forcing separation.* The onset conditions give the coherent forcing a sixth-order zero at the anchor ($\omega_{\text{nom}} \sim \tau^2$, hence $\delta\phi \sim \tau^3$ and $\rho \sim \delta\phi^2 \sim \tau^6$, leading order for $C_2\tau \lesssim 1$), which is exactly where the unstable kernel amplifies most; conversely the forcing reaches $\rho_{\max}$ only at the window edge, where the kernel is unity. At the window midpoint the forcing is below 1% of its edge value. The quoted deviations are therefore *not* obtained from the uniform bound: the exact nominal remainder profile,

$$\rho(s) = \tfrac{1}{2}\, G_2 \left[\tfrac{C_1}{C_2}\big(\sinh C_2 s - C_2 s\big)\right]^2, \qquad G_2 = W(l_p + \lambda_{\text{off}})\,(1 + 5\%_{\text{GE}}), \tag{9}$$

is convolved with the kernel by direct numerical quadrature of (8) — no series truncation; the $\tau^6$ law is only the small-argument limit, and the exact profile is in fact *more* edge-concentrated than the power law, its tail growing exponentially. This exact propagation tightens the uniform bound by a factor 25–96; a hand-derivable version splits the window at its midpoint, where the early half carries less than 1/128 of the forcing mass while the late-half kernel stays below $\cosh(C_2\tau_{\text{end}}/2)$, squeezing the convolution between 1 and ≈2.2 times the plain integral of $\rho$. Since (9) enters *without* first removing its span-absorbable part, the result is doubly conservative. (b) *State dependence.* A priori the remainder is evaluated along the nominal trajectory; the small-gain closure with $L = \max|d\rho/d\phi| \approx 0.95$ N m/rad $\ll W z_{\text{CoM}}$ costs a factor $(1 - L/W z_{\text{CoM}})^{-1} \leq 1.11$. (Along the measured trajectory, Sec. 0.1.3, the bound is unconditional.) (c) *Window length.* Since $\delta\phi(\tau) = (C_1/C_2)(\sinh C_2\tau - C_2\tau)$ is strictly monotone ($d\delta\phi/d\tau = \omega_{\text{nom}} > 0$), the angle threshold determines $\tau_{\text{end}}$ bijectively: $C_2\tau_{\text{end}} \in [2.1, 3.9]$ over the candidate rates, hence $\coth(C_2\tau_{\text{end}}/2) \in [1.04, 1.28]$ and the bound is effectively time-free, $|\delta\omega/\omega| \leq 1.3\, C_2\rho_{\max}/\dot{M}$.

**Design consequence: the excitation cutoff and rate set.** Because the propagated deviation is a computable function of the ramp rate and the excursion cap alone, the ideal analysis *selects* the excitation design. The systematic deviation grows with the cap and is worst at the slowest rate, crossing 1% at 4.8° for 0.10 N m/s and only at 8.9° for 1.20 N m/s; conversely the noise-limited residual floor falls with the cap (13% at 3°, 8% at 5°) and the post-onset sample count exceeds 40 everywhere. The 5° attitude cutoff is therefore the largest common cap that keeps the coherent deviation at the 1% level for every candidate rate, the binding constraint being the slowest ramp. In absolute terms the propagated deviation is 3.1–5.2 mrad/s at the 5° cap and at most 23 mrad/s at 9–10° — an order of magnitude below the in-motion gyro noise (≈ 35 mrad/s RMS).

### 0.1.3 Transfer to the experimental configuration

The experiment departs from the ideal configuration in one respect: the sloped test surface places the rest attitude — and hence the onset, since the vehicle does not rotate before it — at $\phi^* \leq 2.6°$ (median

1.7°, measured over all 140 runs). The correct operating point is $(0, \phi^*, T^*)$, the expansion variable becomes the excursion $\delta\phi = \phi - \phi^*$ (measured: at most 9.4° over the fit window, median 6.1°), and every operating-point effect transfers with an explicit bound:

- the effective anti-restoring gradient shifts from $W z_{\mathrm{CoM}}$ to $W z_{\mathrm{CoM}} \cos\phi^* + W(l_p{+}\lambda_{\mathrm{off}}) \sin\phi^*$ — at most +2.3%, absorbed by the calibrated $C_2$;
- the onset-instant small-angle bias grows from zero to $\frac{1}{2}W(l_p{+}\lambda_{\mathrm{off}})\phi^{*2} \leq 4.9$ mN m per direction — common to both tip directions, which start from the same rest attitude, hence cancelling in $M_{\mathrm{ff}}$;
- the linearization remainder is in fact *smaller* at the tilted point (17.3 vs. 18.5 mN m at the 5° excursion), the restoring term subtracting from the curvature; the design tables, functions of the excursion alone, apply unchanged.

The slope's genuine effect is geometric, not modelling: it renders the two pitch directions asymmetric in runway ($\approx 3°$ vs. 7° to the absolute cutoff), lowering the SNR of the negative-pitch runs — precisely the regime for which the physics-constrained detector is needed.

On the experimental side every component of $\rho$ is also *measured*: the recorded attitude and rotor speeds are evaluated through the exact expressions and projected out of the estimator span $\{1, \tau, \delta\phi\}$ — an algebraic identity, these being the estimator's exact per-run degrees of freedom. Table 2 compares the ideal-theory bounds with the measured values; the measured residuals, on which the claims rest, lie consistently below their a priori counterparts. For the ground effect the projection is evaluated with and without rotor–rotor interference: the measured slope ratio is +0.99% of $\dot{M}$ (interference neglected) and +4.20% (interference included, IQR [4.13, 4.27]%), *both* rate-invariant across the twelvefold rate range — the property the $K$-calibration absorption requires — and the out-of-span residual grows from 0.16 / 2.3 to 0.67 / 9.3 mN m (median RMS / max) with interference, while remaining edge-concentrated, so the propagated shape deviation stays at the few-percent level, below the noise-dominated residual. One term of the affine decomposition deserves explicit accounting. Because the moment coefficient is itself attitude dependent, the product $b(\phi)M_{\mathrm{cmd}}$ contains $\beta_M\, \dot{M}\tau\, \delta\phi$ — a bilinear term in two spanned coordinates, hence the one piece of the ground-effect moment that is *not* in $\{1, \tau, \delta\phi\}$. Evaluated along the measured trajectories with $\beta_M = 0.0345$ rad$^{-1}$ (`analysis/ge_bilinear.py`), its raw peak is 3.0 mN m and what survives the projection is 0.17 mN m at most (median 0.04) — the product is strongly correlated with the spanned coordinates along real trajectories, so it is absorbed almost entirely. It needs no separate treatment: propagated through the same Duhamel integral as every other component — and conservatively, using the *unprojected* forcing — it deforms the rate by 0.04% (slowest ramp) to 0.09% (fastest) of the nominal, against 0.8–1.9% for the small-angle curvature over the same windows. Its smallness has a structural reading: since $\varphi \approx \frac{1}{6}C_1C_2^2\tau^3$ near the onset, the term grows as $\tau^4$ and is therefore edge-concentrated where the kernel has no time left to amplify it; equivalently, $\beta_M\dot{M}\tau$ is a *time-varying increment of the $\varphi$-coefficient*, i.e. the effective anti-restoring gradient drifts by $\beta_M\dot{M}\tau_{\mathrm{end}}/D \leq$ 0.19% across the window — a drift that small produces a response all but indistinguishable from the fixed-gradient cosh, which is why the projection removes 94% of it. As a state-dependent term it also enters the small-gain constant, $\partial\rho/\partial\varphi = \beta_M\dot{M}\tau \leq 0.021$ N m/rad, raising $L$ by 2.2% and leaving the closure factor unchanged to two digits. The channel fates are model-independent: the thrust channel (rotor-common and body lift alike) acts through the arm $l_p$ exactly, is antisymmetric between tip directions, and cancels in $M_{\mathrm{ff}}$ regardless of its magnitude; the moment coefficient leaves a relative bias on $M_{\mathrm{ff}}$ between −1.0% (no interference) and −4.3% (with interference), i.e. 3–14 mN m or 0.10–0.44 mm of CoM offset — below the validation scatter (52 mN m RMS), which is dominated by the genuine contact asymmetries the pivot-free average is designed to capture, and present identically at lift-off, where $M_{\mathrm{ff}}$ compensates it by design. The full chain of reductions is 71.2 → 27.3 → 12.3 → 1.2 mN m: the exact remainder (6) at the largest absolute tilt (10°), the same expression at the median measured excursion (6.1°), the measured pointwise maximum after projection, and the measured window RMS.

### 0.1.4 Error budget: from the residual forcing to millimetres

Table 2 bounds each channel in moment units; the deliverable is a length. The two are joined in closed form, with no free step, so every entry of Table 2 can be read directly as an error of the identified CoM offset.

**Propagation.** The zero onset conditions leave no homogeneous part in (7). The position kernel satisfies $h_\phi(0) = 0$, so the Leibniz boundary term in $\dot{e}_\phi = h_\phi(0)\rho(\tau) + \int_0^\tau \dot{h}_\phi(\tau{-}s)\rho(s)\,ds$ vanishes for *any*

Table 2: Components of the residual forcing $\rho$ over the fit window. "A priori" bounds are the ideal-configuration values of Sec. 0.1.2 evaluated at the measured tilt range; "measured" values are evaluated along the recorded trajectories and projected out of the estimator span $\{1, \tau, \delta\phi\}$. All values in mN m.

| Component | A priori | Measured (RMS / max) | Character |
|---|---|---|---|
| Small-angle curvature | 18.5 at 5°; 27.3 at the median excursion | 1.2 / 12.3 | coherent, edge-concentrated |
| Ground effect, without / with rotor interference | 3.2 / 7.5 (tangent deviation, full range) | 0.16/2.3 → 0.67/9.3 | coherent, edge-concentrated |
| Bilinear $\tau\,\delta\phi$ (GE moment channel) | 3.9 (envelope of the factor maxima) | 0.04 / 0.17 | coherent, almost fully absorbed |
| Ramp nonlinearity | 30 (gate, full window) | 1.6 / 6.2 (fit segment) | incoherent (noise-like) |

forcing — no assumption on $\rho(0)$ is needed — and $e_\phi(0) = \dot{e}_\phi(0) = 0$ then forces both constants of the complementary solution to zero. The deviation is the Duhamel integral (8) alone, evaluated with $J_P = 1/(KC_2^2)$ from the calibrated constants, so no independent inertia estimate enters.

**Translation.** The onset is the argument minimizing the two-segment residual. Perturbing the data by $\delta\omega$, linearizing the stationarity condition about the unperturbed optimum, and using $\partial\hat{\omega}/\partial t_0 = -C_1C_2\sinh(C_2\tau)$ gives

$$\Delta t_0 = -\frac{\langle \delta\omega,\ \sinh C_2\tau\rangle}{C_1C_2\,\|\sinh C_2\tau\|^2}, \qquad \Delta M_{\mathrm{crit}} = \dot{M}\,\Delta t_0 = -\frac{\langle \delta\omega,\ \sinh C_2\tau\rangle}{K\,C_2\,\|\sinh C_2\tau\|^2}, \tag{10}$$

the ramp rate cancelling because $C_1 = K\dot{M}$: the exchange rate from residual forcing to identified threshold is a constant of the calibration, not of the run.

Two identities make this concrete. First, the deviation is not a member of the cosh family: expanding the kernel,

$$\delta\omega(\tau) = \frac{1}{J_P}\Big[P(\tau)\cosh C_2\tau - Q(\tau)\sinh C_2\tau\Big], \qquad \begin{aligned} P(\tau) &= \textstyle\int_0^\tau \cosh(C_2 s)\rho(s)\,ds, \\ Q(\tau) &= \textstyle\int_0^\tau \sinh(C_2 s)\rho(s)\,ds, \end{aligned} \tag{11}$$

whose coefficients drift with $\tau$; they are constant only for $\rho \equiv 0$. This makes the escape channel explicit, and its structure follows from the spanned directions: solving (7) for each spanned forcing shows that the three absorbable directions are exactly the three deformations of the family — $\rho \propto 1$ gives $\delta\omega = \rho_0\sinh(C_2\tau)/(J_PC_2)$, an onset shift; $\rho \propto \tau$ gives $\delta\omega = (c/D)(\cosh C_2\tau - 1)$, an amplitude ($C_1$) change; $\rho \propto \delta\phi$ moves the gradient and hence $C_2$. Since $\rho$ is *defined* as what remains after those are removed, its response has no fixed-coefficient family component.

Second, (10) passes an exact check. Substituting the constant-forcing response above, $\Delta M_{\mathrm{crit}} = -\rho_0/(J_PC_2^2K) = -\rho_0$, since $J_PC_2^2 = D$ and $K = 1/D$: a constant residual moment displaces the identified threshold by exactly itself, which is what the physics requires — the vehicle tips on $\rho_0$ less commanded moment. The translation is therefore unit-faithful, and it bounds the general case: only the projection of $\delta\omega$ on the onset direction $\sinh(C_2\tau)$ biases the result; the rest appears as fit residual, which the NRMSE certificate of Sec. 0.1.5 measures.

**An unconditional bound.** Exchanging the order of integration in (10) reduces the entire propagation to a weighted average of the forcing,

$$\Delta M_{\mathrm{crit}} = -\int_0^{\tau_{\mathrm{end}}} \rho(s)\,w(s)\,ds, \quad w(s) = \frac{\int_s^{\tau_{\mathrm{end}}} \sinh(C_2\tau)\cosh\big(C_2(\tau - s)\big)\,d\tau}{J_PKC_2\,\|\sinh C_2\tau\|^2}, \tag{12}$$

whose kernel is non-negative on the window and *normalized*: $\int_0^{\tau_{\mathrm{end}}} w = 1$ is precisely the constant-forcing identity $\Delta M_{\mathrm{crit}} = -\rho_0$ just established (verified to $5 \times 10^{-4}$ on the discrete operators actually used, `analysis/error_budget.py --check`). Consequently

$$|\Delta M_{\mathrm{crit}}| \;\le\; \max_{[0,\tau_{\mathrm{end}}]} |\rho|$$

holds without further assumptions — no envelope, no coth factor and no small-gain closure are needed, and nothing is assumed about the *shape* of $\rho$, only about its magnitude. Across all 140 runs $\max|\rho| \leq 15.9$ mN m, so no coherent modelling residual of any profile can bias the identified offset by more than 0.50 mm.

The measured value is a further $18\times$ smaller (0.90 mN m, 5.7% of the bound), because $\rho$ changes sign within the window and the weighted mean largely cancels: $|\langle\rho\rangle_w|/\max|\rho|$ is 0.5–1.6% in median by channel and at most 12.6%. The resulting ladder — $15.9 \to 0.90 \to 0.14$ mN m, equivalently 0.50 mm $\to 28\,\mu$m $\to 4.3\,\mu$m — separates cleanly what is bounded by magnitude alone, what the measured sign structure adds, and what the pivot-free pairing removes. (The envelope of Sec. 0.1.2 is not superseded: it bounds the *shape* deviation $\delta\omega/\omega$, which is what the excitation design tables and the NRMSE comparison need, whereas (12) bounds the deliverable directly.)

The deliverable follows the pipeline exactly, $M_{\mathrm{ff}} = \pm\frac{1}{2}(M_+ + M_-)$ and $\lambda_{\mathrm{off}} = M_{\mathrm{ff}}/W$. The estimate is conservative three times over: it forwards the *peak* of each channel, it uses the plain $L^2$ Gauss–Newton curvature whereas the implemented cost is Huber-robust and also refits the pre-onset segment (both stiffen the minimum), and it omits the antisymmetry that removes part of the result at the pairing step.

**Budget.** Table 3 carries every channel of Table 2 through (8) and (10) on all 140 gate-passing runs (`analysis/error_budget.py`). The total coherent modelling error of the identified CoM offset is $1.3\,\mu$m (median) and $4.3\,\mu$m (worst configuration) — roughly $400\times$ below the 1.6–1.8 mm load-cell validation RMS of the identified offset, and $100\times$ below the 0.39 mm that the 3° window ablation moves. Over this dataset the identification is therefore limited by measurement noise rather than by the closed form, which is also what the rate-invariance certificate indicates. The channel ranking is also informative: the gravity curvature dominates and is common-mode, so the pivot-free pairing does not reduce it ($0.99\times$), whereas the ground-effect channels are antisymmetric between tip directions and are partly removed by it ($1.3$–$1.4\times$) — the structural statement of Sec. 0.1.3, now with a number.

Finally, the same table settles how much of $\rho$ could be mistaken for a mis-calibrated $C_2$. Since $\partial \ln(\cosh x - 1)/\partial \ln x = x\coth(x/2) \geq 2$ for all $x > 0$ (and $\in [2.2, 5.3]$ over the measured $C_2\tau_{\mathrm{end}} \in [1.08, 5.20]$), a relative shape deviation $\varepsilon$ can masquerade as at most an $\varepsilon/2$ relative change of the exponent. With the measured $\varepsilon$ (0.26% median, 5.9% worst run) this is $|\Delta C_2/C_2| \leq 0.13\%$ (median) and 3.0% (worst) — below the 12.5% a priori spread the calibration already has to resolve, and in any case unreachable, since $C_2$ is not re-estimated per run: the sole per-run degree of freedom is the onset index, which is exactly what (10) quantifies.

That bound is moreover loose near the anchor, where the coherent model error and an exponent error differ not in size but in *order*. Along the nominal trajectory $\delta\phi \sim \tau^3$, so the lowest-order surviving channel is the bilinear ground-effect term, $\beta_M \dot{M}\tau\,\delta\phi \sim \tau^4$ — the smallest channel overall is the slowest to vanish, the gravity curvature ($\propto \delta\phi^2$) being $\tau^6$. Convolved with $h_\phi(u) \simeq u/J_P$ this gives $e_\phi \sim \tau^6$, i.e. $\delta\omega \sim \tau^5$, against $C_1\,\Delta C_2\,C_2\,\tau^2$ for a genuine exponent error. Equivalently, the forcing that would produce an exact $A(\cosh C_2'\tau - 1)$ response is $\rho = A(J_P C_2'^2 - D)\cosh C_2'\tau + AD$, which requires $\rho(0) = AJ_P C_2'^2 \neq 0$: a forcing alive *at* the onset, which the onset conditions forbid. This, rather than the vanishing of $A_1$ and $A_2$ — a condition the $\cosh -1$ shape itself satisfies — is the structural obstruction, and it confines exponent mimicry to the window edge, where both shapes have relaxed into exponentials and only the amplitude distinguishes them; that is the regime the $\varepsilon/2$ bound covers.

Two qualifications keep this a statement about the coherent model error alone. It is an anchored statement, the absorbable coefficients being matched at $\tau = 0$; the least-squares projection used for the measured residuals instead trades anchor accuracy for window-wide RMS, so $|\rho(0)|$ is not small there (0.17–0.52 of the window peak in median). And near the anchor the measured residual is noise-dominated, its fitted leading power being 1.2–1.7 rather than 4: the record does not resolve the order separation, which is a property of the modelling error and is certified instead by the rate invariance of Sec. 0.1.5. Neither qualification touches Table 3, which forwards the projected *peak* through the Duhamel integral and invokes no onset-order argument.

### 0.1.5 Empirical verification: a prediction test, not a fit

Because the constrained estimator has no free shape parameter per run — $C_1 = K\dot{M}$ with $\dot{M}$ measured, $C_2$ shared, baseline fixed by continuity — the post-onset curve is fully determined before the post-onset data are seen; goodness of fit is evidence, not flexibility. With $\mathrm{NRMSE} = [\frac{1}{N}\sum_k(\omega_k - \hat{\omega}_k)^2]^{1/2}/\max_k|\omega_k - C|$, four measured, model-independent certificates close the claim:

1. *Rate invariance.* Across all 140 gate-passing runs the NRMSE is 6.3% (median; $R^2 = 0.95$) and flat

Table 3: Error budget of the coherent modelling residuals, evaluated on all 140 gate-passing runs. Each channel of Table 2 is propagated through the Duhamel integral (8) and the onset sensitivity (10). "Shape dev." is $|\delta\omega|_{\max}$ relative to the nominal rate at the window edge. The last column is the resulting error of the delivered CoM offset, $\Delta M_{\text{ff}}/W$, after the pivot-free pairing.

| Channel | $\lvert\rho\rvert_{\max}$ [mN m] median / max | Shape dev. [%] median / max | $\lvert\Delta M_{\text{crit}}\rvert$ [mN m] median / max | $\lvert\Delta\lambda_{\text{off}}\rvert$ [$\mu$m] median / max |
|---|---|---|---|---|
| Gravity curvature | 2.75 / 12.3 | 0.10 / 2.17 | 0.037 / 0.300 | 1.14 / 3.74 |
| Ground effect (interf.) | 0.42 / 3.20 | 0.01 / 0.34 | 0.003 / 0.046 | 0.11 / 0.15 |
| Bilinear $\tau\,\delta\phi$ | 0.04 / 0.17 | 0.001 / 0.007 | 0.0005 / 0.007 | 0.01 / 0.03 |
| Ramp nonlinearity | 3.93 / 15.9 | 0.12 / 3.98 | 0.035 / 0.604 | 0.58 / 1.24 |
| **Total** | — | 0.26 / 5.92 | 0.048 / 0.900 | **1.30** / **4.28** |

from 0.10 to 1.20 N m/s (5.5–6.6%): a state-dependent deformation would grow with amplitude, and none does. The residual sits at 2.2× the quiet pre-onset gyro floor — motion- and thrust-induced measurement noise — while the predicted systematic deviation ($\leq 1\%$) lies below it.

2. *Trajectory-evaluated forcing.* The small-angle forcing deviates from the absorbable span by at most 12.3 mN m (median RMS 1.2), edge-concentrated where the kernel provides no amplification; the ground-effect forcing by 2.3 to 9.3 mN m depending on whether rotor interference is included. The measured slope ratios, +0.99% of $\dot{M}$ (without interference) and +4.20% (with), reproduce each model's coefficient in situ and are rate-independent in both cases — the property on which the $K$-calibration absorption rests.

3. *Window ablation.* Truncating the window at a 3° relative excursion — where the small-angle model is unimpeachable — changes the per-direction critical moments by $\leq 0.022$ N m (within scatter) and $M_{\text{ff}}$ by $\leq 0.012$ N m (0.39 mm of CoM offset).

4. *Family exclusivity.* The $d \to 0$ quadratic limit, fitted over the same windows, incurs 14–75% truncation error: the data select the cosh member, not merely any smooth monotone curve.

In summary: structurally, every modeled effect renames the effective constants $(C_2, K)$ that the calibration estimates, so to first order the response cannot leave the cosh family; the sole escape channel is measured — not merely bounded — at a few mN m, and its exact propagation through the deviation dynamics and the onset sensitivity (Table 3) moves the delivered CoM offset by at most $4.3\,\mu$m, some 400× below the achieved accuracy: over this dataset the accuracy is set by measurement noise rather than by the closed form. Empirically, the zero-freedom family predicts the response across all runs with a noise-dominated, rate-invariant residual, is insensitive to the window extent, and decisively outperforms its own quadratic limit. The theoretical bounds serve as conservative a priori envelopes; every headline figure is a measurement.